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Comparative treatment efficiency and water quality index of drinking water systems in North-West Province, South Africa

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Abstract

Access to safe drinking water is a fundamental environmental health requirement, yet integrated assessments of treatment performance and health risks under different land-use pressures remain limited in South Africa. This study evaluated the performance of two drinking water treatment plants in Mahikeng and Klerksdorp, North-West province, representing agricultural and industrial catchments, respectively. A cross-sectional study design was employed. Physicochemical parameters were analysed using the American Public Health Association (APHA) Standard Methods for the Examination of Water and Wastewater, while microbiological quality was assessed by membrane filtration to enumerate *Escherichia coli* and total coliforms. Data were analysed using descriptive statistics and one-way Analysis of Variance (ANOVA), and results were evaluated against South African National Standards (SANS 241) and World Health Organisation (WHO) drinking water guidelines. Both treatment plants achieved complete microbial removal, without detectable *E. coli* or total coliforms, indicating effective disinfection. The water quality index (WQI) values classified treated water as moderate in Mahikeng (70–73) and good in Klerksdorp (30–35). However, physicochemical deviations were identified. Mahikeng recorded elevated turbidity (6.64 ± 9.50 NTU), electrical conductivity (183.50 ± 2.12 $\mu\text{S}/\text{cm}$), total hardness (261.94 ± 15.27 mg/L), and residual chlorine (4.14 ± 0.33 mg/L), while Klerksdorp showed an increase in total dissolved solids (491.50 ± 54.45 mg/L) with lower residual chlorine (1.50 ± 0.37 mg/L). These deviations may reduce treatment efficiency, promote microbial regrowth, and affect water acceptability. Although most of the parameters met the SANS 241 and WHO guidelines, the findings reveal system-level vulnerabilities linked to the quality of source water and the optimisation of the treatment. The study highlights the need for integrated, risk-based monitoring frameworks to ensure sustainable drinking water safety under heterogeneous environmental pressures.

Highlights

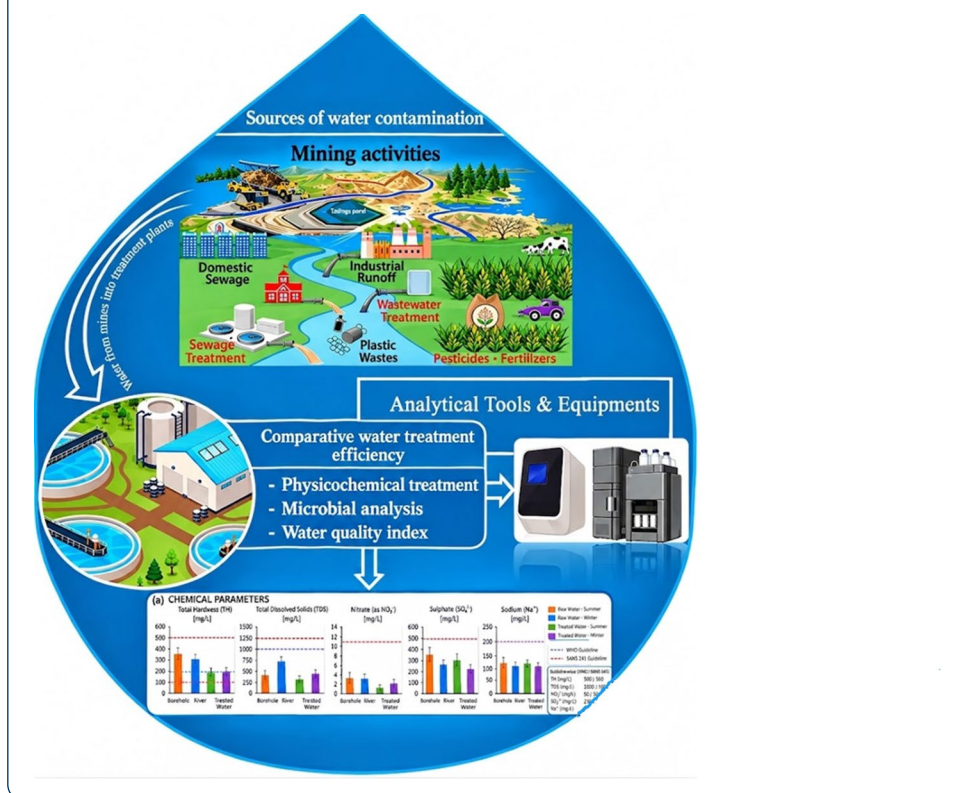
- The Klerksdorp water treatment plant achieved a higher treatment efficiency despite poorer source water quality.
- The treated water from both plants was largely in accordance with WHO guidelines and SANS 241 drinking water standards.



- Seasonal variations had limited influence on the overall stability of treated water quality.
- Persistent turbidity and total hardness exceedances in Mahikeng require process optimisation.
- Improved source water protection and catchment management are essential for sustainable drinking water quality.

Keywords Drinking water quality, Microbial contamination, Water treatment performance, Land-use pressure, Water quality index (WQI)

Graphical abstract



1 Introduction

Water is a fundamental resource for the sustainability of life, and the provision of safe drinking water is critical to protecting public health [1]. Potable water should not pose significant acute or chronic health risks to consumers [9]. The quality of drinking water is evaluated by measuring key physicochemical and microbiological parameters and comparing them with established guideline values, including those set by the World Health Organisation [48] and national regulatory frameworks. The South African Bureau of Standards (SABS) [37] establishes the permissible limits and compliance criteria for domestic water use under the South African National Standards (SANS-241), thereby ensuring that water quality meets prescribed requirements for safety and acceptability [18].

Despite comprehensive regulatory frameworks and international commitments under Sustainable Development Goal 6, the provision of safe drinking water remains a significant global challenge, with approximately one-quarter of the world's population still lacking access to safe managed drinking water services [49]. The Joint Monitoring

Programme (JMP) of the World Health Organisation (WHO) and the United Nations Children's Fund (UNICEF) (2025) reported that approximately 2.2 billion people lacked access to safely managed drinking water services in 2022, highlighting the continuing gap between policy commitments and practical implementation. This challenge is further exacerbated by increased population growth, rapid urbanisation, industrialisation, contaminated agricultural runoff by pesticides, and climate variability, all of which place additional pressure on already limited freshwater resources [30]. To address the increasing pressure on freshwater resources, alternative water management strategies, including wastewater reclamation and greywater reuse, have been widely advocated and increasingly adopted [33]. However, the successful implementation of these approaches depends on effective treatment processes, as inadequately treated reclaimed water can contain biological, chemical, and radiological contaminants that pose potential risks to public health [51].

While microbiological safety is often prioritised, elevated turbidity reduces disinfection efficiency by shielding microorganisms and facilitating pathogen persistence, thereby increasing the risk of waterborne disease transmission [23, 47]. In contrast, elevated turbidity levels are commonly associated with raised concentrations of natural organic matter in water, which may promote the formation of disinfection by-products (DBPs) during chlorination. Additionally, DBPs have also been associated with potential carcinogenic and mutagenic effects in long-term exposure scenarios [25, 48]. Moreover, elevated electrical conductivity indicates increased concentrations of dissolved ions, including sodium, chloride, nitrates, and potentially toxic metals in drinking water. According to the World Health Organisation (2022), prolonged exposure to high concentrations of these dissolved constituents may contribute to adverse health outcomes, including hypertension, kidney dysfunction, and metabolic disturbances, particularly among vulnerable populations.

Total hardness, which is determined primarily by calcium and magnesium, is not inherently toxic; its health implications remain complex and context-dependent [21]. An excessive level of hardness may reduce the acceptability of water and promote scaling within distribution pipelines [24], whereas low hardness (soft water) increases corrosivity, thereby causing a substantial enhancement of the leaching of toxic metals into drinking-water supplies [52]. In general, deviations in physicochemical parameters, including pH and total dissolved solids, can alter the bioavailability of contaminants, indirectly increasing human exposure to harmful substances [36].

In South Africa, municipalities and water boards are mandated under the Water Services Act (Act No. 108 of 1997) to abstract, treat, and distribute potable water to urban, peri-urban, and selected rural communities. To support this mandate and ensure compliance with national drinking water standards, the Department of Water and Sanitation [13] has established comprehensive regulatory monitoring systems, including the Blue Drop Certification Programme and the Integrated Regulatory Information System (IRIS), which provide annual assessments of drinking water quality, treatment plant performance, and regulatory compliance. Nevertheless, the 2023 Blue Drop Report highlights considerable variation in the operational performance, process control, technical capacity, and regulatory compliance of water treatment systems across municipalities, indicating that maintaining consistently safe drinking water remains an ongoing challenge. Consequently, site-specific evaluations of treatment efficiency and drinking water

quality are essential to identify performance gaps and support evidence-based interventions for improving water treatment operations and protecting public health.

The North-West province presents a unique context for assessing the performance of drinking water treatment due to its diverse land-use patterns, encompassing agriculturally dominated and industrialised catchments. For example, the Hartbeespoort Dam, one of the water resources of the province, has experienced persistent eutrophication associated with excessive nutrient enrichment from wastewater effluent, agricultural runoff, and urban activities, resulting in increased algal biomass and declining ecological water quality [15]. Similarly, the Mooi River system, which supports communities and industrial activities in the Klerksdorp–Potchefstroom region, has been affected by elevated salinity and metal contamination associated with mining activities, agricultural practices, and wastewater management challenges, increasing demand for treatment for downstream water purification systems [20]. Mahikeng is characterised by agricultural activities that can influence nutrient and mineral loading in source water, whereas Klerksdorp is influenced by industrial activities that can contribute to elevated dissolved solids and other contaminants. This diversity contributes to considerable variability in raw water quality, as agricultural runoff, industrial discharges, mining activities, and natural geochemical processes can introduce nutrients, dissolved salts, metals, and other contaminants that may increase the complexity of treatment and influence final water quality. These contrasting settings provide an opportunity to comparatively evaluate the performance of the treatment plant in differing environmental settings.

Therefore, this study aims to comparatively assess the performance of domestic water treatment plants in Mahikeng and Klerksdorp by evaluating seasonal variations in key physicochemical and microbiological parameters at both raw and treated water points while examining compliance with regulatory standards. In addition, overall treatment performance was evaluated using the Water Quality Index, a valuable tool to summarise water quality for drinking purposes. Unlike previous studies that focus primarily on compliance, this work explicitly links drinking water treatment efficiency to potential human health risks associated with deviations in key parameters. By examining the performance of drinking water treatment in contrasting agricultural and industrial catchments, the study advances understanding of how the characteristics of the source water influence the outcomes of treatment and residual risk. The findings contribute new empirical evidence to the limited body of knowledge on the operational performance of municipal treatment systems in South Africa and similar developing countries. The results provide a framework for risk-informed water quality management, regulatory decision-making, and targeted optimisation of treatment processes in resource-constrained settings.

2 Materials and methods

2.1 Description of the study area

The present research study was conducted in Mahikeng and Klerksdorp in the North-West province of South Africa (Fig. 1). Mahikeng, the capital of the North-West province, is located within Latitude $-25^{\circ}51'$ S and Longitude $25^{\circ}38'$ E, covering a total land area of 3 698 km². Most of the land in the municipality is farmland (55%) and then traditional area (44%). Less than 2% of the Mahikeng area is urban area [29]. In Mahikeng, the Mahikeng Water Treatment Works (Mahikeng WTW) treats raw water abstracted

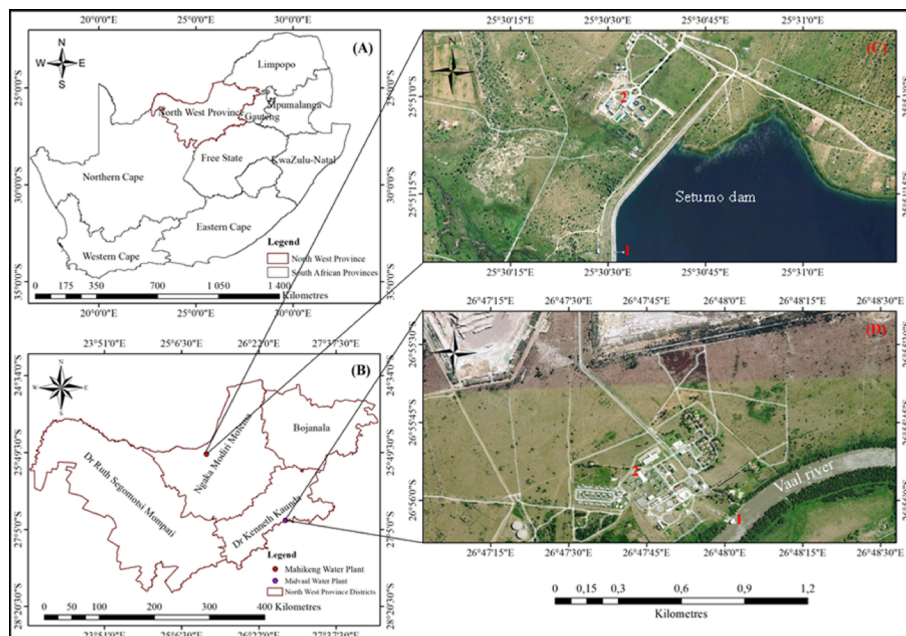


Fig. 1 Map of the North-West province and location of study sites and sampling points

from Setumo dam to produce potable water (Fig. 1C), which is subsequently distributed to consumers within the Mahikeng Local Municipality [31].

The Mahikeng WTW is a conventional surface water treatment facility that supplies potable water to the town of Mahikeng and surrounding communities. In 2016, the plant was upgraded from an initial capacity of 20 ML/day to a design capacity of 30 ML/day to improve the reliability of the bulk water supply in the area [13]. The upgrade incorporated additional treatment infrastructure, including flocculation mixing channels, settling tanks, dissolved air flotation units, and rapid gravity sand filters, enhancing the plant's capacity and treatment performance [14]. Based on the plant's design capacity and typical municipal domestic water consumption rates, the facility has the potential to supply approximately 150,000–200,000 people and Mahikeng has an estimated population of 354 504 [41].

Klerksdorp is located at approximately 26°52' S latitude and 26°40' E longitude and encompasses an area of approximately 105.98 km². The town is characterised by heterogeneous and intensive land-use patterns, shaped predominantly by mining, agricultural activities, commercial and industrial zones, and urban settlements [44]. The Midvaal Water Company (MWC) Water Treatment Plant (hereafter referred to as Klerksdorp Water Treatment Plant WTP), located near Klerksdorp in the North-West province, is a major bulk potable water supplier serving the City of Matlosana, including Klerksdorp, Orkney, Stilfontein and Hartbeesfontein, as well as mining and industrial consumers [44]. The plant abstracts water from the Vaal River (Fig. 1D) and has a design treatment capacity of approximately 320 ML/day, although operational assessments have indicated that it currently operates below its design capacity, with an assessed operating capacity of approximately 130 ML/day and a potential optimisation capacity of approximately 250 ML/day. The treatment process consists of a conventional multi-barrier surface water treatment system incorporating coagulation, flocculation, dissolved air flotation, clari-flocculation, filtration, ozonation, chlorination, and clear water storage. Based on

the plant's design capacity and typical municipal domestic water consumption rates, the facility has the potential to supply approximately 150,000–200,000 people. The local municipality of the City of Matlosana has an estimated population of 431 231 [41].

The study areas have a semi-arid climate characterised by warm to hot summers (which usually span from October to April), cool dry winters (which usually span from May to September every year), and an average annual rainfall of around 482 mm to 633 mm, mostly falling in summer. The average temperatures during summer range from 21 to 32 °C in Mahikeng, while in Klerksdorp the average temperature ranges from 18 to 30 °C.

2.2 Materials

2.2.1 Workflow

The methodological workflow used in this study is presented in Fig. 2, highlighting the sequential steps used for sample collection, laboratory analyses, determination of the Water Quality Index (WQI), statistical evaluation, and assessment of the drinking water treatment performance.

2.2.2 Water sampling, physicochemical testing and data processing

Water sampling

Grab water samples were collected at the influent (source water) and treated water (effluent) after the completion of the treatment process, allowing the evaluation of the treatment performance and compliance with the drinking water quality standards. The collection and analysis of water samples was conducted bi-weekly during two sampling campaigns, in June 2018 (lower temperatures) and October 2018 (higher temperatures), to capture seasonal variation in physicochemical and microbiological parameters [32]. Unless otherwise specified by the guidelines of the former Department of Water Affairs and Forestry (DWAF), now the Department of Water and Sanitation (DWS), ultrapure deionised water was used for the preparation and dilution of all standards and samples [10, 12]. All field equipment and laboratory instruments were calibrated according to the manufacturer's instructions, and analyses were performed following established standard operating procedures for water quality assessment.

Six batches were collected per study site in triplicate using high-density polyethylene (HDPE) bottles (1000 mL) for physicochemical analyses. The bottles were washed with laboratory detergent, thoroughly rinsed with tap water followed by deionised water, soaked in 10% nitric acid for 24 h, rinsed with ultrapure deionised water, and air-dried before use. Before sampling, bottles were rinsed three times with sample water, except where preservation requirements prohibited rinsing [10]. Four containers were allocated for physicochemical analyses and required no pre-treatment. Sterile polypropylene sampling bottles used for microbiological analyses were autoclaved at 121 °C for 15 min before use and pre-dosed with sterile 10% $\text{Na}_2\text{S}_2\text{O}_3$ solution (0.1 mL per 100 mL sample) to neutralise residual chlorine and prevent disinfection during transport [3].

The sampling points at Mahikeng WTW are illustrated in Fig. 3, while Fig. 4 shows the sampling points at Klerksdorp WTP. Water samples were taken at the inlet (1) to determine the physicochemical and biological status of water before the treatment process, and the outlet (2) to determine the physicochemical and biological status of water after the treatment process.

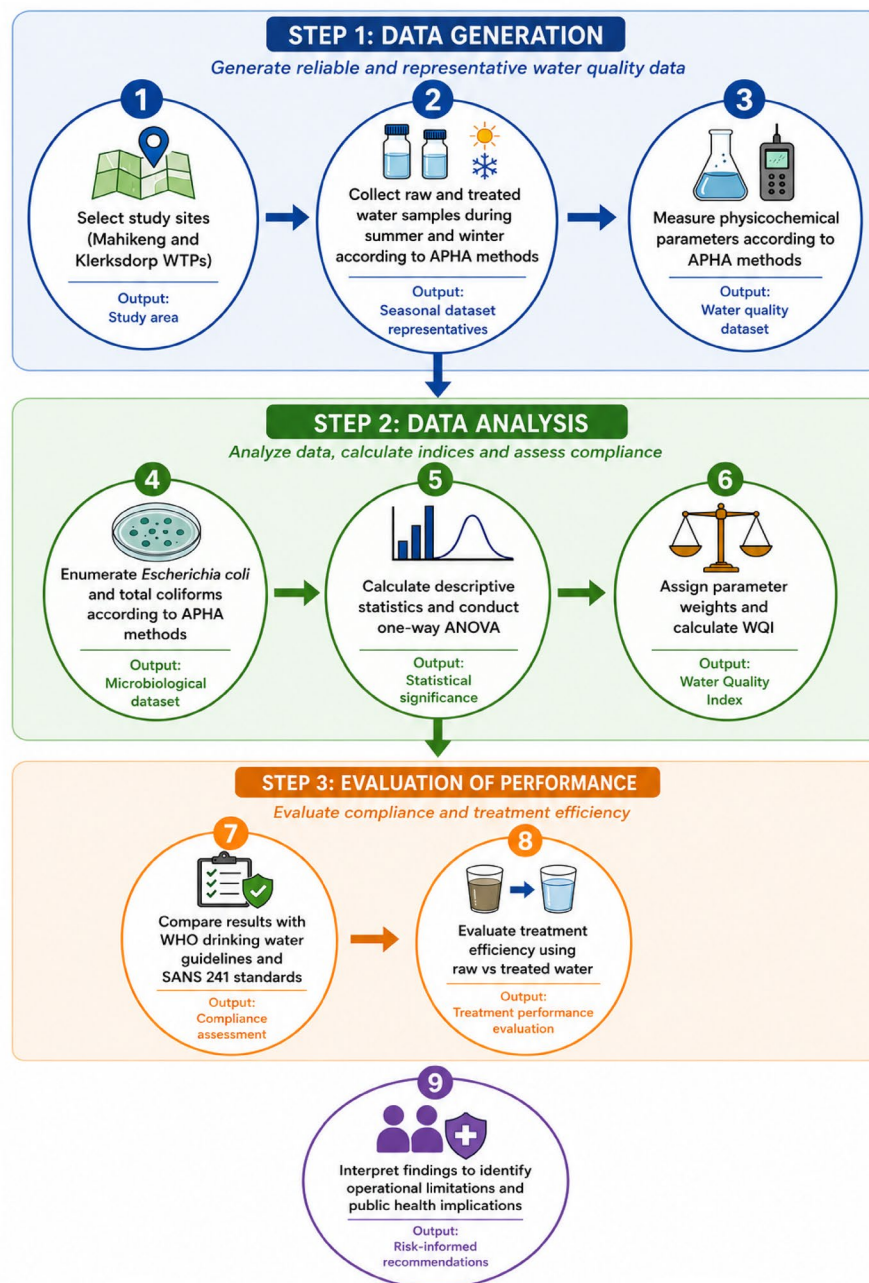


Fig. 2 Integrated framework for evaluating drinking water treatment performance

The collected samples were kept in an ice cooler and transported to the North-West University Chemistry Laboratory for further analysis.

2.2.3 Description of the dataset

The sampling, preservation, transportation, and laboratory analyses were conducted in accordance with the Department of Water Affairs (DWAF) sampling protocols, which adopt the analytical procedures outlined in the APHA Standard Methods for the Examination of Water and Wastewater [3] and the South African Bureau of Standards [34] to promote the reliability and reproducibility of the data. The datasets comprised 12 variables of drinking water quality, including ten physicochemical parameters (temperature,

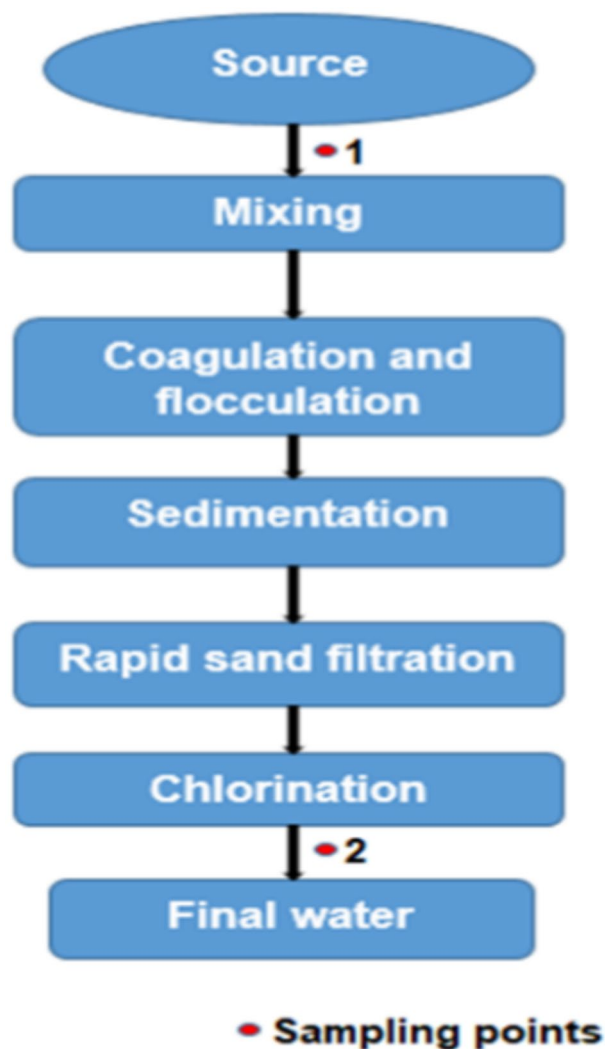


Fig. 3 Flow chart showing the sequence of the conventional water treatment steps and the sampling points used at Mahikeng WTW

pH, electrical conductivity, turbidity, residual chlorine, total hardness, total dissolved solids, nitrate, sulphate and sodium) and two microbiological indicators (*Escherichia coli* and total coliforms). These variables were chosen to reflect their relevance to the evaluation of treatment performance, regulatory adherence, and possible public health hazards.

Physicochemical analyses were performed within the recommended holding times, while microbiological analyses were performed within 6 h after collection, consistent with the APHA recommendations [3]. Datasets were further examined using descriptive statistics, one-way Analysis of Variance (ANOVA), and the evaluation of Water Quality Index (WQI) to determine seasonal variation, treatment efficiency, and alignment with drinking water quality standards.

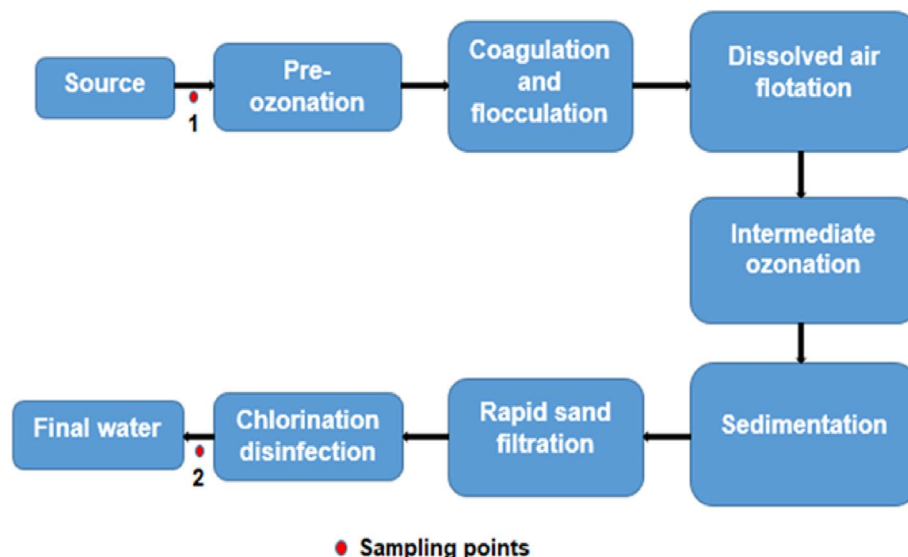


Fig. 4 Flow chart showing the sequence of the water treatment steps, plus the sampling points at the Klerksdorp WTP

Table 1 Water quality parameters and analytical methods for water samples evaluation

Parameter	APHA standard method (23rd ed., 2017)	Analytical method	Instrument
pH	4500-H ⁺ B	Electrometric method	Portable pH/ORP meter; Hach South Africa (Pty) Ltd, Roodepoort, Gauteng
Temperature (°C)	2550 B	Electrometric (thermometer)	Temperature probe integrated into pH meter; Hach South Africa (Pty) Ltd, Roodepoort, Gauteng
EC	2510 B	Conductometric method	Conductivity meter; Hach South Africa (Pty) Ltd, Roodepoort, Gauteng
Turbidity	2130 B	Nephelometric method	IR turbidimeter; Lasec Group, Ndabeni, Cape Town
Residual Cl ₂	4500-Cl G	DPD colorimetric method	Free chlorine DPD test kit and colorimeter; Hach South Africa (Pty) Ltd, Roodepoort, Gauteng
TH	2340 C	EDTA titration	n/a
TDS	2540 C	Gravimetric method	n/a
NO ₃ ⁻	4500-NO ₃ ⁻ E	UV Spectrophotometric	Spectroquant Prove 300; Merck KGaA, Darmstadt, Germany
SO ₄ ²⁻	4500-SO ₄ ²⁻ E	UV Spectrophotometric	Spectroquant Prove 300; Merck KGaA, Darmstadt, Germany
Na ⁺	3500-Na B	Flame emission spectrophotometry	Flame spectrophotometer; Hanna Instruments Inc., Woonsocket, Rhode Island

2.3 Analysis of water samples

2.3.1 Physicochemical parameters

Table 1 summarises the physicochemical water quality parameters analysed, and the corresponding analytical methods employed. Detailed description of the calibration and analytical procedures appear in the supplementary material. For determination of total hardness, the study used the EDTA titrimetric method according to the APHA Standard Method 2340 C [3]. A 50 mL aliquot of each water sample was buffered to pH 10 using an NH₄Cl buffer, followed by the addition of the Eriochrome Black T indicator. The samples were titrated with a standardised 0.01 M EDTA solution until the endpoint colour

changed from red to clear blue. The total hardness was calculated using Eq. (1) reported as mg/L CaCO_3 .

$$\left[\text{Total hardness} \left(\frac{\text{mg}}{\text{L}} \text{ as } \text{CaCO}_3 \right) \right] = \frac{50 \times 1000 \times M_B \times V_B}{V_A} \quad (1)$$

where: 50 = the equivalent weight of calcium carbonate (CaCO_3) in mg; 1000 = conversion from mL to L; M_B = EDTA concentration (mg/L); V_B = volume of EDTA consumed (mL); V_A = sample volume (mL).

2.3.2 Microbiological analysis

The enumeration of *Escherichia coli* (*E. coli*) and total coliforms was carried out using the membrane filtration technique according to APHA Standard Method 9222 [3]. The samples (50 mL) were filtered through sterile 0.45 μm membrane filters (manufactured by Derbgum, Johannesburg, South Africa) using a Gelman Little Giant pressure/vacuum pump machine (model 13156, Gelman Sciences, Ann Arbor, MI, USA). Membrane filters were aseptically transferred onto m-Endo agar and m-FC agar (Merck, Darmstadt, Germany) to selectively isolate total coliform and faecal coliform bacteria, respectively. The membranes were placed onto m-Endo agar LES for total coliforms and mFC agar for *E. coli*, followed by incubation at $37 \pm 0.5^\circ\text{C}$ and $44.5 \pm 0.2^\circ\text{C}$, respectively, for 24 h. Presumptive colonies were confirmed using Gram staining, oxidase, citrate utilisation, triple sugar iron tests, and the EnteroPluri-Test (Ref: 78618–78619) [34].

2.3.3 Water quality control and assurance

Water quality control and assurance procedures were implemented to ensure the accuracy, reliability, and reproducibility of the analytical results. The sampling was conducted using standard protocols to minimise contamination and variability during collection, preservation, transport, and analysis. Field parameters, including pH, electrical conductivity, temperature, and turbidity, were measured using calibrated portable instruments, while laboratory analyses were performed according to the APHA Standard Methods for the Examination of Water and Wastewater [3]. Quality assurance measures included instrument calibration, analysis of reagent blanks, duplicate samples, and quality control standards to verify analytical precision and accuracy. Data validation involved reviewing analytical results for consistency, assessing replicate measurements, and confirming compliance with quality control criteria prior to statistical analysis and Water Quality Index (WQI) calculations. The reliability of the dataset was further ensured by comparing the measured water quality parameters with the limits specified in the South African National Standard (SANS 241: Drinking Water Specification) [37] and the Domestic Drinking Water guidelines of the World Health Organisation [48]. SANS 241 was selected as the primary benchmark because it represents the legally recognised national standard for the quality of potable water in South Africa and defines acceptable limits for microbiological, chemical, physical and aesthetic parameters. The WHO guidelines were used as an international reference framework to provide a broader context and facilitate comparison with internationally accepted drinking water quality criteria. The use of both standards allowed for an assessment of treatment performance, regulatory compliance, and the general suitability of treated water for human consumption.

2.4 Data analysis

2.4.1 Statistical analysis

(i) Descriptive statistics

Data for all physicochemical parameters measured in the water samples were recorded and analysed using Microsoft Excel 2013. The mean and standard deviation (SD) were calculated for each sampling site and sampling point.

(ii) Significance testing

A two-tailed significance test ($P < 0.05$) was applied to assess the differences between seasons and sampling sites.

(iii) Treatment efficiency

The treatment efficiency of each water treatment plant was evaluated based on the percentage of removal of selected physicochemical parameters between raw and treated water. The removal efficiency (%) was calculated using the difference between influent and effluent concentrations according to Eq. 2:

$$\text{Treatment efficiency (\%)} = \frac{C_{\text{raw}} - C_{\text{treated}}}{C_{\text{raw}}} \times 100 \quad (2)$$

where C_{raw} = the concentration of a parameter in raw water; C_{treated} = concentration of a parameter after treatment.

The calculated removal efficiencies for physicochemical parameters were used to compare the performance of the Mahikeng WTW and Klerksdorp WTP treatment systems. This approach enabled evaluation of the effectiveness of the treatment process in reducing contaminants and improving treated water quality relative to South African (SANS 241) standards and the international (WHO) drinking water quality guidelines.

2.4.2 Assessment of the water quality index

The Water Quality Index (WQI) is a composite metric that integrates multiple water quality parameters into a single value, facilitating straightforward interpretation by water resource managers [40]. The WQI was calculated in three steps. First, each of the nine parameters was assigned a weight (w_i) based on its relative importance to drinking water quality. For this study, the guidelines recommended by the World Health Organisation [48] were used to calculate WQI.

Second, a maximum weight of 5 was assigned to parameters of the greatest significance, while a minimum weight of 1 was assigned to those with comparatively minor influence. Intermediate weights (1–5) were assigned according to the relative contribution of each parameter to overall water quality. The WQI was calculated from the raw water and treated water results using the weighted arithmetic index method [3] based on Eq. 3.

$$\text{Relative Weight (} W_i \text{)} = \frac{w_i}{\sum_{i=1}^n w_i} \quad (3)$$

where (W_i) represents the weight per parameter, n is the number of parameters, and W_i is the relative weight. Weight (w_i) calculated relative weight (W_i).

The third step was to calculate the quality scale (q_i) for each parameter using Eqs. 4 and 5 for pH:

$$q_i = \frac{C_i}{S_i} \times 100 \quad (4)$$

For pH:

$$q_i = \frac{C_i - 7.0}{8.5 - 7.0} \times 100 \quad (5)$$

where (q_i) is the quality ranking, (C_i) is the concentration of the parameter in each water sample (mg/L), and (S_i) is the WHO guideline for the parameter (mg/L).

To compute the WQI, Eq. 6 was used to determine the sub-index (SI_i) of each parameter and thereafter, Eq. 7 was used to calculate the WQI.

$$SI_i = \frac{\sum_{i=1}^n W_i q_i}{\sum_{i=1}^n W_i} \quad (6)$$

$$WQI = \sum SI_i \quad (7)$$

where (SI_i) is the sub-index of the i th parameter, q_i is the rating based on the concentration of the i th parameter, W_i is the unit weight assigned to the i th parameter, and n is the total number of parameters.

The calculated WQI values are classified into five categories (Table 2) [4].

3 Results and discussion

3.1 Physical water quality

The comparative physicochemical characteristics of the water samples from the study area and their corresponding World Health Organisation and South African National Standards allowable limits are presented in Figs. 5 and 6. Table 3 shows the seasonal mean values for the physical parameters from the two study sites.

The pH of treated water from the Klerksdorp WTP remained within the recommended WHO guidelines and the standard limits of SANS 241 throughout the sampling period, ranging between 7.68 and 8.24 in summer and winter, respectively (Fig. 5). In comparison, the pH of Mahikeng WTW treated water ranged from 8.29 to 9.33 during winter and summer, respectively (Fig. 6). These values were within the recommended limits for potable water as stipulated by the World Health Organisation and SANS 241. Although both water treatment plants effectively maintained pH within acceptable limits, the elevated summer pH values observed in Mahikeng may be of concern, as pH values above 8 can adversely affect the efficiency of disinfection processes in water treatment systems [17].

Table 2 Water quality based on WQI using weighted arithmetic water quality index (WAWQI) method

WQI range	Type of water
< 50	Excellent water
50–100	Good water
100–200	Poor water
200–300	Very poor water
> 300	Water unsuitable for drinking purposes

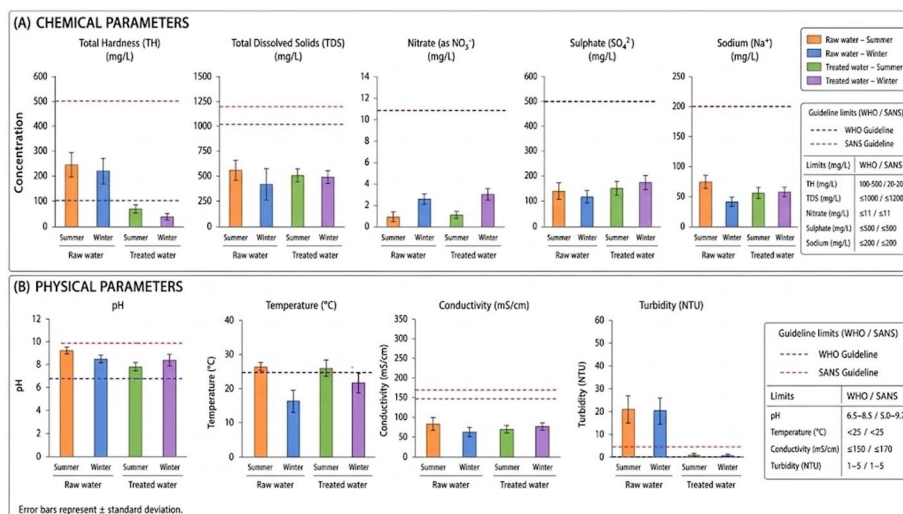


Fig. 5 Physicochemical water quality parameters from Klerksdorp WTP

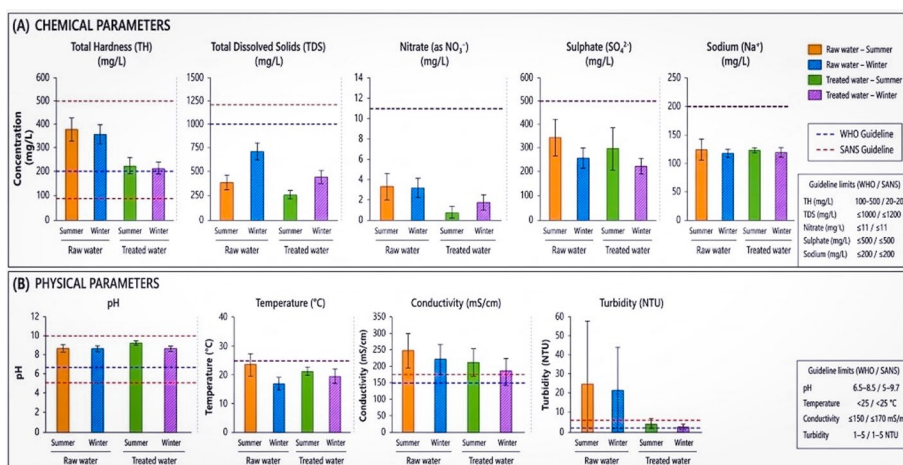


Fig. 6 Physicochemical water quality parameters from Mahikeng WTW

Temperature is a key parameter that influences the physical, chemical, and biological properties of water and consequently the overall chemistry of water. It is inherently variable in natural water bodies due to fluctuating environmental conditions [28]. Following treatment, the mean temperatures recorded at the sampling sites met the recommended guideline limits of $<25^{\circ}\text{C}$ as specified by WHO and SANS 241. Although a slight exceedance of the allowed limit was observed during the summer season in Klerksdorp, no statistically significant differences were detected between pre- and post-treatment temperatures ($P = 0.43$).

The post-treatment electrical conductivity values of the Klerksdorp WTP of $69.50 \pm 2.12 \mu\text{S/cm}$ (summer) and $78.00 \pm 7.07 \mu\text{S/cm}$ (winter) were all within the prescribed drinking water quality guidelines, demonstrating effective treatment performance. In contrast, the post-treatment conductivity values in Mahikeng WTW remained above the recommended limits prescribed by WHO and SANS 241, with mean values of $203.00 \pm 16.97 \mu\text{S/cm}$ in summer and $183.50 \pm 2.12 \mu\text{S/cm}$ in winter, indicating limited efficiency of the treatment process in reducing dissolved ions. Electrical conductivity is a

Table 3 Seasonal mean values for physical parameters and free chlorine

	pH		Temperature (°C)		Conductivity (µS/cm)		Turbidity (NTU)		Free chlorine (mg/L)	
	Summer	Winter	Summer	Winter	Summer	Winter	Summer	Winter	Summer	Winter
MWTP 1	8.91 ± 0.05	8.36 ± 0.08	26.00 ± 0.14	15.58 ± 2.56	240.00 ± 76.37	204.50 ± 36.06	23.43 ± 100.37	23.34 ± 34.97	0	0
MWTP 2	9.33 ± 0.01	8.29 ± 0.06	25.65 ± 1.48	21.94 ± 2.50	203.00 ± 16.97	183.50 ± 2.12	5.37 ± 16.92	6.64 ± 9.50	3.60 ± 0.18	4.14 ± 0.33
MWC 1	9.32 ± 0.11	8.39 ± 0.47	22.86 ± 3.51	18.90 ± 0.88	83.50 ± 7.78	60.50 ± 26.16	21.50 ± 1.65	20.67 ± 0.94	0	0
MWC 2	7.68 ± 0.15	8.24 ± 0.16	21.88 ± 0.54	20.79 ± 1.88	69.50 ± 2.12	78.00 ± 7.07	0.53 ± 0.03	0.39 ± 0.12	1.47 ± 0.01	1.50 ± 0.37
WHO	6.5 ≥ to ≤ 8.5		No guideline		≤ 150		1 ≥ to ≤ 5		0.2 ≥ to ≤ 5	
SANS	5.0 ≥ to ≤ 9.7		No guideline		≤ 170		1 ≥ to ≤ 5		0.2 ≥ to ≤ 5	

NB: MWTP = Mahikeng WTP; MWC = Klerksdorp WTP

1 represents inlet values, and 2 represents outlet values

key indicator of water purity, reflecting the concentration of dissolved ions and the ability of water to conduct an electric current [38]. Elevated conductivity values can indicate high salinity or potential pollution, while low values are typically associated with low-quality, low-nutrient waters [7]. These findings indicate that the Mahikeng treatment process had limited effectiveness in reducing dissolved ions, suggesting that the existing treatment system is not designed to remove high concentrations of dissolved salts. Elevated conductivity reflects an increase in ionic content, which may originate from natural geological formations or anthropogenic activities within the catchment. In addition, these findings highlight the need for improved source water management and the incorporation of treatment technologies specifically targeting dissolved salts where necessary.

Long-term consumption of water with elevated electrical conductivity (EC), such as that observed in Mahikeng WTW, is not necessarily harmful, but it reflects a high concentration of dissolved ions that may have indirect health implications depending on their composition. Previous studies have shown that prolonged consumption of highly saline drinking water, as indicated by elevated electrical conductivity, can contribute to increased blood pressure, hypertension, and potentially a greater risk of cardiovascular disease, particularly in populations exposed to high sodium concentrations in drinking water [16]. In some cases, the elevated levels of SO_4^{2-} and other salts can also cause chronic gastrointestinal irritation or a laxative effect in some consumers [48].

Turbidity is a key indicator of water clarity that is widely used to assess drinking water quality because it reflects the presence of suspended particles, including silt, clay, organic matter, and microorganisms. Elevated turbidity has both health and aesthetic implications, as it can interfere with disinfection processes, facilitate microbial persistence, and reduce the acceptability of drinking water by consumers [50]. In Klerksdorp WTP, elevated turbidity levels were observed at the inlet, ranging from 21.50 ± 1.65 NTU in summer to 20.67 ± 0.94 NTU in winter. Following treatment, turbidity levels were substantially reduced to 0.53 ± 0.03 NTU and 0.39 ± 0.12 NTU for summer and winter, respectively (Fig. 5), falling within the recommended WHO guidelines and the SANS 241 drinking water standards, indicating an effective treatment process. This reduction was statistically significant ($p < 0.001$). In contrast, Mahikeng WTW exhibited markedly high levels of turbidity at the inlet, with mean values of 23.34 ± 100.37 NTU in summer and 23.43 ± 34.97 NTU in winter, exceeding the allowed limits of 1–5 NTU. These high levels of turbidity can be attributed to seasonal rainfall-induced runoff, catchment erosion, agricultural and urban runoff, and the influx of organic matter into the source water [39]. Although treatment reduced turbidity to 5.37 ± 16.92 NTU and 6.64 ± 9.50 NTU for summer and winter, respectively (Fig. 6), these values remained above acceptable standards, suggesting suboptimal treatment efficiency and the potential inadequacy of the disinfection process to remove particulate matter and associated microbial contaminants. A statistically significant difference was also observed between the levels of turbidity at the inlet and outlet of the Mahikeng WTW ($P = 0.008$). Similar findings were reported by [27], who found that turbidity at two rural water treatment plants and several points within the distribution system in the Vhembe District frequently exceeded both the aesthetic guideline of 5 NTU and the operational target of 1 NTU. The authors attributed the elevated turbidity to operational and maintenance challenges, noting that high turbidity can compromise disinfection efficiency by shielding microorganisms from chlorine exposure. In contrast, Bwapwa [5] reported that the Hazelmere Water

Treatment Plant in KwaZulu-Natal Province, South Africa, consistently achieved treated water turbidity of ≤ 1 NTU, with turbidity removal efficiencies ranging from approximately 66% to 99.6%, demonstrating that well-operated conventional treatment processes can effectively produce water that complies with WHO guidelines and SANS 241 standards.

The elevated levels of turbidity observed in Mahikeng WTW are of considerable public health concern because turbidity serves as an indicator of suspended particulate matter that can harbour and protect pathogenic microorganisms from disinfection [47, 50]. Consequently, the associated health implication is due to the potential persistence of microbial contaminants rather than to turbidity itself [45]. Elevated turbidity can reduce the effectiveness of disinfection processes by shielding microbes, thus increasing the likelihood of waterborne diseases caused by bacteria, viruses, and protozoa [23]. This can result in acute gastrointestinal illnesses, particularly among vulnerable populations. Additionally, the presence of organic matter in turbid water can promote the formation of harmful disinfection by-products during chlorination, which are linked to long-term health effects including carcinogenic and reproductive risks [25, 48]. Therefore, the elevated post-treatment turbidity observed in the present study suggests that optimisation of coagulation, flocculation, sedimentation, and filtration processes may be required to improve treatment performance and ensure more effective reduction of microbial risk.

3.2 Chemical indicators of water quality

Table 4 shows the seasonal mean chemical parameters from the two study sites.

Total hardness (TH) represents the combined concentration of calcium and magnesium ions in water [22]. Drinking water is classified as soft (< 61 mg/L as CaCO_3), moderately hard (61–120 mg/L as CaCO_3), hard (121–180 mg/L as CaCO_3), and very hard (> 180 mg/L as CaCO_3) [11]. After treatment, only the TH values for Klerksdorp WTP were reduced to within acceptable limits, decreasing from 240.83 ± 46.22 mg/L to 67.81 ± 20.98 mg/L in summer, and from 216.58 ± 76.09 mg/L to 42.56 ± 11.76 mg/L in winter (Fig. 5), indicating effective treatment performance. In contrast, TH values for Mahikeng WTW remained elevated at 267.64 ± 13.56 mg/L in summer and 261.94 ± 15.27 mg/L in winter (Fig. 6), exceeding the limits of the WHO guidelines and SANS 241 standards and suggesting suboptimal treatment efficiency. The difference in TH concentrations between study areas could be attributed to the natural geology of the areas [42]. The northern region of the North-West province, where Mahikeng is located, is rich in limestone deposits. Ca^{2+} , the main cation that contributes to the hardness of the water, may leach into groundwater and surface water as it percolates through these limestone formations, particularly during the rainy summer season [46].

Consumption of water with elevated hardness is generally not considered a direct health hazard; however, excessive TH may have indirect implications for human health. Mirzabeygi [29] elaborated that high concentrations of calcium and magnesium can contribute to scaling in domestic systems. Kozisek (2020) noted that elevated total hardness (TH) can reduce the palatability and general consumer acceptability of drinking water. Although hardness is generally not considered a direct health concern, the consumption of elevated hardness of water has been associated with gastrointestinal discomfort in susceptible individuals. In addition, elevated water hardness together with a high level

Table 4 Seasonal mean values for chemical parameters

	Sulphate (mg/L)		Sodium (mg/L)		TH (mg/L)		TDS (mg/L)		Nitrate (mg/L)	
	Summer	Winter	Summer	Winter	Summer	Winter	Summer	Winter	Summer	Winter
MWTP 1	168.51 ± 24.61	131.75 ± 83.44	133.05 ± 3.48	127.71 ± 1.31	360.52 ± 4.88	352.36 ± 19.76	230.53 ± 11.43	326.24 ± 53.97	3.07 ± 0.23	2.24 ± 1.37
MWTP 2	187.97 ± 29.32	183.05 ± 6.99	129.40 ± 0.14	124.56 ± 5.76	267.64 ± 13.56	261.94 ± 15.27	166.64 ± 9.50	214.34 ± 16.92	0.83 ± 0.82	1.61 ± 0.88
MWC 1	135.53 ± 16.01	123.42 ± 83.76	75.00 ± 6.36	42.69 ± 24.48	240.83 ± 46.22	216.58 ± 76.09	525.50 ± 51.62	376.30 ± 143.54	0.71 ± 0.66	2.31 ± 0.96
MWC 2	152.23 ± 23.29	174.60 ± 23.31	56.89 ± 1.97	59.30 ± 5.94	67.81 ± 20.98	42.56 ± 11.76	491.50 ± 54.45	472.50 ± 50.20	0.82 ± 0.81	2.72 ± 0.25
WHO	≤ 500		≤ 200		≥ 100 to ≤ 500		≤ 1000		≤ 11	
SANS	≤ 500		≤ 200		≥ 20 to ≤ 200		≤ 1200		≤ 11	

NB: MWTP = Mahikeng WTP; MWC = Klerksdorp WTP
1 represents inlet values, and 2 represents outlet values

of TDS may contribute to an increased dietary intake of minerals that could pose risks to individuals with certain renal conditions or those prone to kidney stone formation [21].

The total concentrations of dissolved solids (TDS) of Klerksdorp WTP exhibited a limited reduction in TDS after treatment, with values decreasing marginally from 525.50 ± 51.62 mg/L (summer) and 376.30 ± 143.54 mg/L (winter) at the inlet to 491.50 ± 54.45 mg/L and 472.50 ± 50.20 mg/L at the outlet (Fig. 5). In contrast, from Mahikeng WTW, TDS decreased after treatment with mean values reduced from 326.24 ± 53.97 mg/L and 230.53 ± 11.43 mg/L at the inlet to 214.34 ± 16.92 mg/L and 166.64 ± 9.50 mg/L at the outlet during winter and summer, respectively (Fig. 6), indicating moderate treatment effectiveness even though the reduction was not statistically significant ($P=0.22$). The observed differences in treatment efficiency can be attributed to the nature of the source water. Mahikeng is predominantly influenced by agricultural activities, whereas Klerksdorp, as an industrial and commercial urban centre, is more likely to be affected by complex dissolved constituents originating from industrial effluents, urban runoff, and commercial activities. These dissolved ions and salts are often not completely removed by conventional water treatment processes, which primarily target suspended solids and microbial contaminants rather than dissolved substances. Consequently, slightly elevated TDS concentrations in Klerksdorp may reflect the nature of its source water rather than treatment inefficiency. Nevertheless, TDS concentrations at both sites remained well within the permissible limits recommended by WHO and SANS 241 for drinking water, indicating that the treated water was aesthetically acceptable and suitable for human consumption. These findings are consistent with those reported by [27], who found that the quality of the source water at two rural water treatment plants in the Vhembe District was influenced by catchment activities. Despite these impacts, the treated water complied with national drinking water standards for physicochemical parameters, including TDS, demonstrating the effectiveness of conventional treatment processes in producing potable water. TDS is not typically associated with direct acute health effects; however, elevated levels may pose indirect health risks depending on the composition of dissolved components. Elevated TDS can contribute to an excessive intake of ions such as sodium and sulphates, which can be associated with stomach irritation, and long-term use can cause hypertension or gastrointestinal irritation [19], while also indicating the possible presence of harmful contaminants [35].

The results in Figs. 5 and 6 show that the concentrations of NO_3^- , Na^+ and SO_4^{2-} at both Mahikeng and Klerksdorp remained within their respective recommended drinking water limits at all sampling points and seasons, indicating that there is no immediate health risk and generally effective treatment processes, despite minor post-treatment increases observed in Klerksdorp. The residual (free) chlorine levels observed in the present study remained within the acceptable range (0.2–5 mg/L) of the values of the drinking water guidelines in both treatment plants, suggesting that the chlorination process was effective despite differences in the quality of the source water. Although Mahikeng recorded comparatively higher residual chlorine concentrations, these values still complied with drinking water standards and likely reflect a higher dose due to poorer source water quality. Comparable results were documented by Maphanga [26], who reported that maintaining a residual chlorine concentration of approximately 0.5 mg/L throughout the distribution network required relatively high initial chlorine doses due to the poor quality of raw water and chlorine decay. The authors concluded

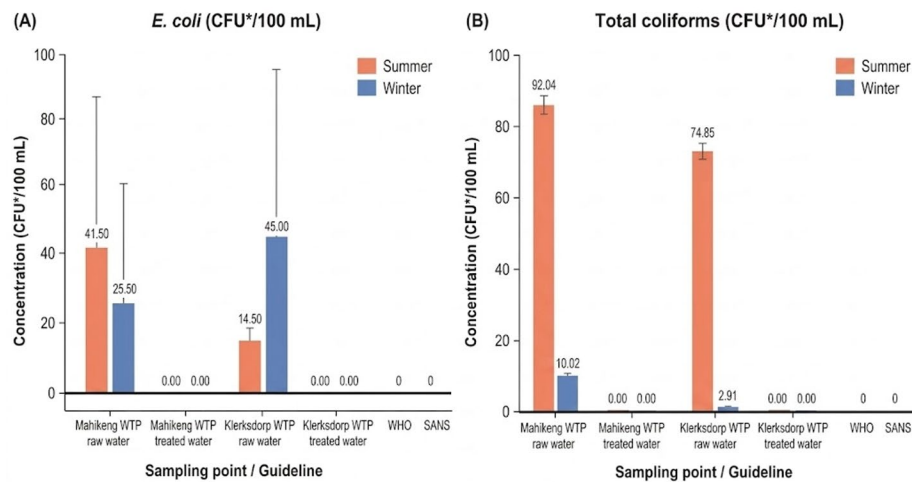


Fig. 7 Seasonal variation of **A** *E. coli* and **B** Total coliforms

Table 5 Seasonal mean values of *E. coli*, and Total coliform

	<i>E. coli</i> (CFU*/100 mL)		Total coliforms (CFU*/100 mL)	
	Summer	Winter	Summer	Winter
MWTP 1	41.50 ± 44.55	25.50 ± 34.65	920356.00 ± 56.57	10024.50 ± 21.93
MWTP 2	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
MWC 1	14.50 ± 4.95	45.00 ± 49.50	74855.00 ± 3047.63	2912.80 ± 2493.05
MWC 2	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
WHO	0		0	
SANS	0		0	

NB: MWTP = Mahikeng WTP; MWC = Klerksdorp WTP

1 represents inlet values, and 2 represents outlet values

that increased chlorine dosing is often necessary to achieve adequate residual disinfectant levels without compromising microbial safety. In general, there were no significant differences in these chemical parameters between the study areas or seasons, supporting the conclusion that the treated water met the regulatory requirements.

3.3 Microbiological parameters

The *E. coli* and total coliform values for both Klerksdorp WTP and the Mahikeng WTP post-treatment were within the WHO and SANS 241 recommended limits, indicating compliance with microbiological drinking water standards (Fig. 7A and B; Table 5).

The results further show that the raw water sources that supply both sites are highly contaminated with bacteria, particularly during the summer season. Overall, the treatment processes were effective in reducing microbial contamination to acceptable levels before distribution. This reduction can be attributed to the integrated performance of conventional treatment barriers, including coagulation–flocculation, sedimentation, filtration, and disinfection. Coagulation and sedimentation promote the removal of particulate matter and particle-associated microorganisms, while filtration and disinfection are the primary barriers to the presence of harmful pathogens in drinking water [43]. The final disinfection step provides microbial inactivation, ensuring that treated water complies with microbiological quality requirements before distribution.

Table 6 Weight and relative weight of each parameter used for WQI calculation

Parameter	WHO standard (S_i)	Weight (w_i)	Relative weight (W_i)
pH	8.5	3	0.091
TH (mg/L)	500	2	0.061
TDS (mg/L)	1000	5	0.152
EC (μ S/cm)	150	4	0.121
Turbidity (NTU)	5	5	0.152
NO_3^- (mg/L)	11	2	0.061
Free Cl_2	5	3	0.091
SO_4^{2-} (mg/L)	500	5	0.152
Na^+ (mg/L)	200	4	0.121
		33	1.000

Table 7 Comparison of raw and treated water WQI

Treatment plant	Season	Raw water WQI	Treated water WQI	WQI Reduction (%)	Treatment performance
Klerksdorp WTP	Summer	105.9 (poor)	30.8 (excellent)	70.9	Strong improvement
Klerksdorp WTP	Winter	91.9 (good)	35.8 (excellent)	61.1	Strong improvement
Mahikeng WTW	Summer	124.7 (poor)	73.1 (good)	41.4	Moderate improvement
Mahikeng WTW	Winter	117.7 (poor)	70.7 (good)	39.9	Moderate improvement

NB: Water quality classification appears in parentheses

3.4 Water quality index

The WHO guidelines for drinking water quality were used for the calculation of WQI. The rating of the water was computed using the physicochemical parameters obtained. Table 6 presents the weights assigned to each parameter for the calculation of the water quality index (WQI) based on their relative importance to the quality of drinking water. Total dissolved solids, turbidity, and sulphate received the highest relative weights (0.152), indicating their greater influence on WQI, while conductivity and sodium were assigned moderate weights (0.121). In contrast, pH, free chlorine, total hardness, and nitrate were assigned lower relative weights, reflecting their comparatively smaller contribution to the overall index. The use of WHO guideline values as reference standards ensures a reliable and internationally recognised assessment of drinking water quality.

The WQI was calculated to evaluate the suitability of treated water for human consumption, with results presented in Table 7. Both plants received raw water classified as 'poor,' with Mahikeng WTW exhibiting the poorest raw water quality (WQI >120) during summer.

The overall performance of the treatment varied between the plants: Klerksdorp WTP showed a substantial improvement, with WQI reductions of 70.9% in summer and 61.1% in winter. In contrast, Mahikeng WTW demonstrated moderate improvement, potentially due to the poorer quality of its raw water and the ageing infrastructure limiting treatment efficiency. Similar trends have been reported in South Africa, where treatment plants receiving poor-quality source water often experience reduced treatment efficiency due to increased contaminant loads and operational challenges [27]. In particular, municipal water systems affected by ageing infrastructure and inadequate maintenance have shown greater difficulty in consistently achieving optimal treated water quality [14].

Table 8 Comparison of treatment efficiencies (%) for selected physicochemical parameters at Mahikeng WTW and Klerksdorp water treatment plants

Parameter	Klerksdorp		Mahikeng WTW	
	% TE (summer)	% TE (winter)	% TE (summer)	% TE (winter)
pH	17.6	1.8	– 4.7	0.8
Temperature (°C)	1.3	– 40.8	4.3	– 10.0
TH (mg/L)	71.8	80.3	25.8	25.7
TDS (mg/L)	6.5	– 25.6	27.7	34.3
Conductivity (µS/cm)	16.8	– 28.9	15.4	10.3
Turbidity (NTU)	97.5	98.1	77.1	71.6
Nitrate (mg/L)	– 15.5	– 17.7	73.0	28.1
Sulphate (mg/L)	– 12.3	– 41.5	– 11.5	– 38.9
Sodium (mg/L)	24.1	– 38.9	2.7	2.5

NB: TE = Treatment Efficiency

3.5 Treatment efficiency

Comparison of treatment efficiencies revealed distinct performance patterns between the two water treatment plants (Table 8). Mahikeng WTW consistently achieved moderate to high removal of turbidity (71.6–77.1%), total dissolved solids (27.7–34.3%), total hardness (25.7–25.8%), and nitrate (28.1–73.0%), with relatively stable treatment efficiencies across seasons. In contrast, the lower removal of TDS and nitrate at Mahikeng is consistent with previous studies demonstrating that conventional treatment processes have limited capacity to remove dissolved inorganic constituents without advanced treatment technologies [2, 6].

Alternatively, Klerksdorp exhibited exceptional turbidity (97.5–98.1%) and total hardness (71.8–80.3%) removal but showed marked seasonal variability, with negative treatment efficiencies for total dissolved solids, conductivity, nitrate, sulphate, sodium, and temperature during winter. Similar seasonal variations in drinking water treatment performance have been attributed to changes in raw water quality, temperature, coagulant demand and operational conditions [8]. The turbidity removal efficiencies observed at Klerksdorp (97.5–98.1%) are comparable to the 95–99% removal efficiencies reported for optimised conventional drinking water treatment plants [8]. Overall, Klerksdorp was more effective at removing particulate contaminants, whereas Mahikeng displayed more consistent performance and greater stability in reducing dissolved constituents across seasons, suggesting differences in source water characteristics and treatment operations.

4 Sustainable development goals alignment and future perspectives

This study directly advances SDG 6 (Clean Water and Sanitation) by evaluating the efficiency of drinking water treatment and identifying opportunities to improve water quality management in South Africa. The findings also contribute to SDG 3 (Good Health and Well-being) through the assessment of residual public health risks, while supporting SDGs 11, 12, 13 and 15 by informing resilient urban water infrastructure, sustainable treatment practices, climate-adaptive water management and integrated catchment protection. This integrated approach strengthens evidence-based strategies for achieving safe, sustainable, and resilient drinking water systems.

Future research should conduct health risk assessments associated with drinking water exposure by integrating contaminant concentrations, consumption patterns, exposure duration, and vulnerable population groups to estimate potential non-carcinogenic and carcinogenic risks. Further research should also investigate the specific

sources contributing to elevated turbidity and total hardness in Mahikeng through detailed catchment assessments, including land-use analysis, sediment characterisation, and identification of both anthropogenic and natural sources.

5 Conclusion and recommendations

5.1 Conclusions

Effective drinking water treatment is essential for protecting public health, particularly in regions where source water quality is influenced by anthropogenic activities and natural variability. This study demonstrated that, despite receiving comparatively more impacted source water, the Klerksdorp treatment system outperformed Mahikeng WTW, consistently producing higher-quality treated water. The Water Quality Index (WQI) classified Klerksdorp water as good (30–35) and Mahikeng water as moderate (70–73). Although the treated water of both plants generally complied with WHO guidelines and the SANS 241 drinking water standard, persistent exceedances in turbidity and total hardness indicate the need for targeted process optimisation and routine monitoring to improve treatment efficiency and distribution system performance. These findings provide valuable evidence for water resource managers and policymakers to prioritise treatment upgrades, strengthen operational controls, and implement site-specific interventions to ensure sustainable delivery of safe and acceptable drinking water.

5.2 Recommendations

The study recommends the following:

- Implement continuous and systematic monitoring of treated water quality to ensure a consistent supply of potable water that complies with national and international regulatory standards.
- Upgrade and modernise existing water treatment infrastructure, particularly ageing treatment systems, to improve operational efficiency, enhance treatment reliability, and reduce the risk of water quality failures.
- Optimise treatment processes to address persistent turbidity and total hardness exceedances, especially in Mahikeng, through improved coagulation–flocculation, filtration and hardness control strategies to improve consumer acceptability.
- Strengthen catchment management practices to reduce the entry of contaminants from anthropogenic activities, including agricultural practices and mining. Improved catchment protection would reduce raw water contamination and lower the treatment burden on water purification systems.

Supplementary Information

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Supplementary Material 1.

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Author contributions

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Data availability

All data supporting the findings of this study are available within the paper and its Supplementary Information.

Declarations

Ethics approval and consent to participate

No human or animal elements were involved in this study. However, the Research and Ethics Committee of the North-West University granted ethical approval for data collection (Ref: NWU-00287-18-A9). Furthermore, the authors have complied with the statement on “Ethical responsibilities of Authors” found in the Instructions for Authors. Not applicable.

Consent for publication

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Competing interests

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